MIND SCOPE: PREDICTION OF MENTAL HEALTH USING MACHINE LEARNING TECHNIQUES

A project report submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI & ML)

by

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CERTIFICATE

This is to certify that the project work entitled, **"MIND SCOPE: PREDICTION OF MENTAL HEALTH USING MACHINE LEARNING TECHNIQUES"** is a bonafide record of the project work done and submitted by

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under my guidance and supervision for the partial fulfillment of the requirements for the award of B.Tech degree in **COMPUTER SCIENCE AND ENGINEERING (AI & ML).** This project is the result of our own effort and that it has not been submitted to any other University or Institution for the award of any degree or diploma other than specified above

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

We hereby declare that the project entitled **"MIND SCOPE: PREDICTION OF MENTAL HEALTH USING MACHINE LEARNING TECHNIQUES",** done by us under the esteemed guidance of **Dr. K. Shivakumaraswamy,** Professor dept.of.CSE(AI&ML) and is submitted in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology in Computer Science and Engineering (AI & ML).**

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ABSTRACT

Mental health issues such as anxiety, depression, and stress have become increasingly prevalent due to lifestyle changes, academic pressure, and social factors. Early detection and timely intervention are vital for improving overall wellbeing and treatment outcomes.

Mind Scope: Mental Health Prediction Using Machine Learning aims to develop an Intelligent system capable of predicting the likelihood of mental health conditions based on user data such as demographics, lifestyle habits, academic or work stress levels, and self-reported symptoms. The system uses the Naive Bayes algorithm, a probabilistic classifier known for its simplicity, interpretability, and effectiveness in handling limited datasets.

The model analyzes input data to estimate the probability of different mental health states, providing early alerts for at-risk individuals. Evaluation metrics such as accuracy, precision, recall, and F1-score are employed to validate the model's performance. By offering an Automate performance. By offering an automated and data-driven approach, Mind Scope assists psychologists, counselors, and healthcare organizations in early identification and prevention of mental health issues.

This project demonstrates how artificial intelligence can support mental health awareness by combining efficiency, interpretability, and accessibility to promote proactive psychological well-being.

CONTENTS

	Abstract	**i**
	Table of Contents	**ii**
	List of figures	**iv**
	List of Tables	**v**
	Abbreviations	**vi**

Chapter No.	Description	Page No.
1	**INTRODUCTION**	**1-2**
2	**PROJECT DESCRIPTION**	**3-5**
	2.1 Problem Definition	3
	2.2 Project Details	4-5
	2.3 Scope of the Project	5-6
3	**SYSTEM ANALYSIS**	**7-10**
	3.1 Existing System	7
	3.1.1 Disadvantages	8
	3.2 Proposed System	9
	3.2.1 Advantages	10
4	**ENVIRONMENTS**	**11-13**
	4.1 Software Requirements.	11
	4.2 Hardware Requirements	11
	4.3 Software Features and Dependencies	12-13
5	**SYSTEM DESIGN**	**14-29**
	5.1 System Architecture	14-15
	5.2 UML Diagrams	16-17
	5.2.1 Class Diagram	17-19
	5.2.2 Use Case Diagram	19-21
	5.2.3 Sequence Diagram	22-23

Chapter No.	Description	Page No.
	5.2.4 Activity Diagram	23-25
	5.2.5 Component Diagram	25-27
	5.2.6 Deployment Diagram	28-29
6	**SYSTEM IMPLEMENTATION**	**30-34**
	6.1 Implementation Process	30-31
	6.2 Module Explanation	32-34
7	**SYSTEM STUDY AND TESTING**	**35-42**
	7.1 Feasibility Study	35-36
	7.2 Software Testing	36-42
8	**SOURCE CODE**	**43-60**
9	**SCREEN LAYOUTS**	**61-63**
10	**FUTURE ENHANCEMENTS**	**64**
11	**CONCLUSION**	**65**
12	**BIBLIOGRAPHY**	**66-68**

LIST OF FIGURES

Fig. No.	Figure Name	Page No.
5.1	System Architecture	15
5.2.1	Class diagram	19
5.2.2	Use Case diagram	21
5.2.3	Sequence diagram	23
5.2.4	Activity diagram	25
5.2.5	Component diagram	27
5.2.6	Deployment diagram	29
6.2.1	List of Modules	32
9.1	Interface Layout	61
9.1.1	Home page	61
9.1.2	Input interface	62
9.1.3	Input data	62
9.2.1	Output 1	63
9.2.2	Output 2	63

LIST OF TABLES

Table No.	Table Name	Page No.
7.2.1	System Validation And Testing Results	37
7.2.2	Detailed Test Case Execution Log	38-39

ABBREVIATIONS

AI	-	Artificial Intelligence
ML	-	Machine Learning
NB	-	Naive Bayes
CSV	-	Comma Separated Values
API	-	Application Programming Interface
GUI	-	Graphical User Interface
UI	-	User Interface
CPU	-	Central Processing Unit
UML	-	Unified Modeling Language
DB	-	Database
EDA	-	Exploratory Data Analysis
F1-SCORE	-	Harmonic Mean of Precision and Recall
JSON	-	JavaScript Object Notation
UX	-	User Experience
NB Classifier	-	Naive Bayes Classifier
NumPy	-	Numerical Python Library
Pandas		Python Data Analysis Library
Sklearn	-	Scikit-learn Machine Learning Library
Joblib	-	Python Library for Model Serialization
Streamlit	-	Python Framework for Web Applications

Chapter 1

INTRODUCTION

Mental health has become a significant concern in recent years, particularly among students, due to increasing academic pressure, rapid lifestyle changes, and evolving social influences. Issues such as stress, anxiety, and depression can adversely affect a student's academic performance, decision-making ability, and overall well-being. In many cases, these conditions remain unnoticed until they reach a severe stage, making early identification and timely intervention crucial for maintaining a healthy and productive life.

Traditional approaches to mental health assessment typically involve manual evaluation methods such as questionnaires, surveys, and expert analysis. Although these methods provide valuable insights, they are often time-consuming, subjective, and limited in accessibility. Furthermore, they may not always offer real-time analysis or scalable solutions for a large population. With the rapid advancement of technology, machine learning techniques have emerged as powerful tools capable of analyzing complex datasets, identifying hidden patterns, and providing predictive insights with improved efficiency and accuracy.

The project titled **"Mind Scope: Prediction of Mental Health Using Machine Learning Techniques"** focuses on developing an intelligent system that can predict mental health conditions among students. The system utilizes the **Naive Bayes algorithm**, a probabilistic classification technique, to analyze various influencing factors such as anxiety levels, self-esteem, lifestyle habits, academic pressure, and social environment. Based on these inputs, the model predicts the likelihood of stress and related mental health conditions, enabling early detection and awareness.

Naive Bayes is widely preferred due to its simplicity, fast computation, and effectiveness in handling structured data. It is based on Bayes' theorem and assumes independence among input features, which allows the model to perform well even with limited training

data. This makes it a suitable choice for real-world applications where data availability and processing time are critical considerations.

The proposed system is implemented as a user-friendly web application that allows users to input their personal and behavioral data through an interactive interface. The system processes this information and provides predictions along with meaningful suggestions or recommendations. By integrating machine learning with an accessible platform, this project aims to support students, educators, and counselors in understanding mental health conditions better and taking proactive measures to improve overall well-being.

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Chapter 2

PROJECT DESCRIPTION

2.1 PROBLEM DEFINITION:

Mental health issues such as stress, anxiety, and depression are increasingly prevalent among students due to academic pressure, lifestyle changes, and social influences. Despite their growing impact, these conditions often go unnoticed or are identified only at later stages, leading to serious consequences affecting both personal and academic life.

Existing methods for assessing mental health primarily rely on manual techniques such as surveys, questionnaires, and consultations with mental health professionals. While these methods can provide useful insights, they suffer from several limitations. They are often time-consuming, subjective in nature, and not easily accessible to all individuals. Additionally, many students hesitate to seek help due to stigma or lack of awareness, further delaying proper diagnosis and support.

There is a clear need for an efficient, accessible, and data-driven system that can assist in the early detection of mental health conditions. Such a system should be capable of analyzing multiple influencing factors, including lifestyle habits, emotional state, academic pressure, and social environment, to provide reliable predictions.

Therefore, the problem addressed in this project is the development of a machine learning-based mental health prediction system that can accurately identify students at risk. The system aims to overcome the limitations of traditional methods by providing a fast, objective, and user-friendly solution for early mental health assessment and awareness.

2.2 PROJECT DETAILS

This project, titled **"Mind Scope: Prediction of Mental Health Using Machine Learning Techniques,"** aims to develop an intelligent and user-friendly system for predicting mental health conditions among students using machine learning techniques. The system focuses on identifying stress levels at an early stage by analyzing various influencing factors such as health conditions, lifestyle habits, environmental factors, and social interactions.

The system begins with user interaction through a **Streamlit-based web application**, where users register and provide input data using intuitive components such as sliders and dropdown menus. The input includes parameters like anxiety level, sleep quality, self-esteem, study load, social support, and other behavioral indicators.

The collected data undergoes preprocessing, where it is converted into a structured numerical format suitable for machine learning models. This step includes data cleaning, feature formatting, and validation to ensure accurate predictions.

The processed data is then passed to the **Naive Bayes classification model**, which is trained to predict the stress level of the user. The model classifies the output into categories such as **Low, Medium, or High stress**, along with a confidence score. The use of Naive Bayes ensures fast and efficient prediction, making it suitable for real-time applications.

The system then displays the prediction results through a visually interactive interface, including a stress meter and percentage score for better understanding. Additionally, the system provides personalized recommendations such as exercise, meditation, and lifestyle improvements to help users manage their mental health effectively.

An analytics module is also integrated into the system, which provides insights into the dataset, including average stress levels and other statistical information. This helps in better understanding overall trends and patterns.

By automating the mental health prediction process, the system reduces dependency on manual assessment methods, enables early detection, and promotes awareness. The solution can be further extended to educational institutions, counseling platforms, and healthcare systems to support proactive mental health management.

2.3 SCOPE OF THE PROJECT

The scope of the project "Mind Scope: Prediction of Mental Health Using Machine Learning Techniques" is focused on developing a web-based application that predicts the mental health status of users, particularly students, using machine learning algorithms. The system is designed to analyze multiple factors such as anxiety levels, sleep quality, self-esteem, academic pressure, lifestyle habits, and social interactions to determine the user's stress level.

The application provides an interactive and user-friendly interface built using Streamlit, where users can input their data through sliders and dropdown menus. Based on the provided inputs, the system processes the data and uses the Naive Bayes model to classify the stress level into categories such as low, medium, or high. The system also displays the results using visual indicators like a stress meter and percentage score, making it easy for users to understand their mental health status.

In addition to prediction, the system offers personalized recommendations such as maintaining a healthy lifestyle, improving sleep patterns, engaging in social activities, and practicing relaxation techniques. An analytics module is also included to provide insights into dataset trends and overall stress levels, enhancing the understanding of mental health patterns.

The project is mainly intended for students, educators, and counselors as a supportive tool for early detection and awareness of mental health issues. It can be deployed as a

standalone web application and can also be integrated into educational platforms or mental health support systems.

However, the scope of the project is limited to predictive analysis and awareness. It does not replace professional medical diagnosis or psychological treatment. The system depends on user-provided data, and the accuracy of predictions may vary based on the quality and completeness of inputs.

Chapter 3
SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Current mental health assessment systems primarily rely on traditional methods such as questionnaires, surveys, and manual evaluation by psychologists or counselors. These methods involve collecting self-reported data from individuals and analyzing it based on predefined psychological frameworks. Although these approaches are widely accepted and clinically relevant, they are often time-consuming, subjective, and not easily accessible to a large population.

In recent years, some semi-automated systems have been introduced using basic machine learning techniques and behavioral data such as smartphone usage, sleep patterns, and social interaction metrics. While these systems attempt to improve prediction accuracy, they still depend heavily on continuous data collection and often require external devices or applications, making them less practical for everyday use.

Advanced models such as the **I-HOPE (Interpretable Hierarchical Model for Personalized Mental Health Prediction)** have shown improvements in prediction accuracy by incorporating hierarchical modeling and personalized behavioral data. However, these systems still face limitations such as handling missing or inconsistent data, computational complexity, and difficulty in real-time deployment.

Additionally, many existing systems focus on generalized or population-level analysis rather than individual-level prediction. This reduces their effectiveness in providing personalized mental health insights. The lack of a simple, user-friendly, and real-time prediction system limits accessibility for students and individuals who may require immediate awareness of their mental health condition.

Overall, current systems lack scalability, accessibility, and real-time responsiveness, highlighting the need for a more efficient, data-driven, and user-centric solution.

3.1.1 Disadvantages

- ***Time-Consuming Process:***
  Traditional mental health assessment methods require significant time for data collection, evaluation, and expert analysis.

- ***Subjective Results:***
  Dependence on self-reported data and human interpretation can lead to inconsistencies and biased outcomes.

- ***Limited Accessibility:***
  Professional mental health services are not easily accessible to all individuals, especially for early-stage detection.

- ***Dependency on Continuous Data:***
  Some advanced systems require continuous tracking through devices, making them less practical for general use.

- ***Lack of Personalization:***
  Many systems provide generalized results rather than individual-specific predictions.

- ***Low User Engagement and Awareness:***
  Many traditional and semi-automated systems are not designed with user-friendly interfaces, leading to reduced user interaction and awareness. This limits their effectiveness in encouraging individuals to regularly monitor their mental health.

3.2 PROPOSED SYSTEM

The proposed system, titled **"Mind Scope: Prediction of Mental Health Using Machine Learning Techniques,"** aims to develop an intelligent, data-driven solution for predicting mental health conditions among students. The system is designed to provide early detection of stress levels by analyzing multiple influencing factors such as anxiety, sleep quality, self-esteem, academic pressure, lifestyle habits, and social interactions.

The system is implemented as a **web-based application using Streamlit**, providing a simple and interactive interface for users. It begins with user registration, followed by data input through sliders and dropdown menus, ensuring ease of use and accessibility for non-technical users.

The input data undergoes preprocessing, where it is converted into a structured numerical format suitable for machine learning. This includes data cleaning, formatting, and feature preparation to ensure compatibility with the prediction model.

The core of the system is the **Naive Bayes algorithm**, which is used to classify the user's mental health status into categories such as **Low, Medium, or High stress levels**. The model is trained on relevant datasets and provides fast and efficient predictions with good accuracy, even when working with limited data.

Once the prediction is generated, the system displays the results through a clear and visually interactive interface, including a stress level indicator and percentage score. Additionally, the system provides **personalized recommendations** such as lifestyle improvements, relaxation techniques, and wellness activities to help users manage their mental health.

An analytics component is also included to provide insights into overall trends and data patterns. By integrating machine learning with a user-friendly platform, the proposed system ensures accessibility, real-time prediction, and improved awareness of mental health conditions.

3.2.1 Advantages

- ***Early Detection:***
  The system helps identify mental health issues at an early stage, enabling timely intervention.

- ***Fast and Efficient:***
  The use of the Naive Bayes algorithm ensures quick predictions suitable for real-time applications.

- ***User-Friendly Interface:***
  The Streamlit-based application provides an intuitive and interactive experience for users.

- ***Improved Accessibility:***
  The system can be accessed easily without requiring professional consultation or specialized tools.

- ***Personalized Recommendations:***
  Provides tailored suggestions to improve mental well-being based on prediction results.

- ***Scalable Solution:***
  Can be extended and integrated into educational institutions and healthcare systems.

- ***Reduced Human Dependency:***
  Minimizes reliance on manual assessment and subjective decision-making.

- ***Privacy-Oriented Usage:***
  Users can access the system independently, reducing hesitation in sharing sensitive information and encouraging honest input.

Chapter 4

ENVIRONMENTS

4.1 SOFTWARE REQUIREMENTS

- Operating System : Windows 10 / Windows 11
- Programming Language : Python
- Front-End : Streamlit (Python-based UI)
- Back-End : Python (Machine Learning Model – Naive Bayes)
- Libraries Used : Pandas, NumPy, Scikit-learn, Matplotlib, Joblib
- IDE/Editor : Visual Studio Code / Jupyter Notebook

4.2 HARDWARE REQUIREMENTS

- Processor : Intel i3 / i5 or above
- RAM : Minimum 4 GB (8 GB recommended)
- Storage : At least 20 GB free disk space
- Input Devices : Standard Keyboard and Mouse
- Display : LCD/LED Monitor

4.3 SOFTWARE FEATURES AND DEPENDENCIES

Python provides a rich ecosystem of libraries and frameworks that support machine learning, data processing, and application development. In this project, relevant Python libraries are used to build an efficient and user-friendly mental health prediction system. The following are the key software dependencies used in the system.

Scikit-learn

Scikit-learn is one of the most widely used machine learning libraries in Python. It provides simple and efficient tools for data analysis and model building. In this project, Scikit-learn is used to implement the **Naive Bayes algorithm** for predicting mental health conditions.

Key Features:

- Provides classification algorithms such as Naive Bayes
- Easy model training and testing
- Includes evaluation metrics like accuracy, precision, recall, and F1-score
- Efficient and suitable for structured datasets

Pandas

Pandas is a powerful data manipulation library used for handling structured data. It plays a crucial role in data preprocessing and preparation.

Key Features:

- Handles data in tabular format (DataFrames)
- Supports data cleaning and transformation
- Efficient handling of missing values
- Easy integration with machine learning models

NumPy

NumPy is a fundamental library for numerical computations in Python. It is used for handling arrays and performing mathematical operations efficiently.

Key Features:

- Supports multi-dimensional arrays
- Fast numerical computations
- Used in data preprocessing and model input preparation

Matplotlib

Matplotlib is a data visualization library used to create graphs and charts. In this project, it is used for analytics and visualization of results.

Key Features:

- Generates plots and charts
- Helps in visual representation of data
- Useful for analyzing patterns and trends

Streamlit

Streamlit is a Python framework used to build interactive web applications easily. It is used to develop the user interface of the system.

Key Features:

- Simple and fast UI development
- Supports real-time user interaction
- Provides widgets like sliders, buttons, and dropdowns
- Ideal for machine learning applications

Joblib

Joblib is used for saving and loading trained machine learning models efficiently.

Key Features:

- Fast model serialization
- Efficient handling of large data objects
- Enables reuse of trained models

Chapter 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The system architecture of the proposed **Mind Scope** application follows a structured pipeline approach that enables efficient data flow from user input to prediction output. The architecture consists of five major components: User, Streamlit User Interface, Application Logic, Machine Learning Model, and Prediction Output.

The process begins with the **User**, who interacts with the system by registering, entering input data, and viewing the results. The user provides details related to mental health factors such as anxiety, sleep quality, lifestyle habits, and social conditions.

The input is then handled by the **Streamlit User Interface (app.py)**, which acts as the front-end of the system. It includes a registration form for basic details like name and age, as well as an input data form using sliders and dropdowns. The interface also provides sidebar navigation for easy movement across different sections of the application.

Once the data is entered, it is passed to the **Application Logic layer (Python)**. This component is responsible for session management, input validation, data preprocessing, and feature formatting. It converts user input into a structured numerical format suitable for the machine learning model.

The processed data is then forwarded to the **Machine Learning Model**, which is based on the Naive Bayes algorithm. The trained model (stored as a .pkl file and loaded using Joblib) analyzes the input data and predicts the user's stress level efficiently.

Finally, the prediction results are displayed in the **Prediction Output module**, where the system shows the results clearly through a visual interface. The output includes stress

level classification and other indicators, helping users easily understand their mental health status.

Overall, the architecture ensures a smooth, real-time, and user-friendly workflow from data input to result generation, making the system efficient and scalable.

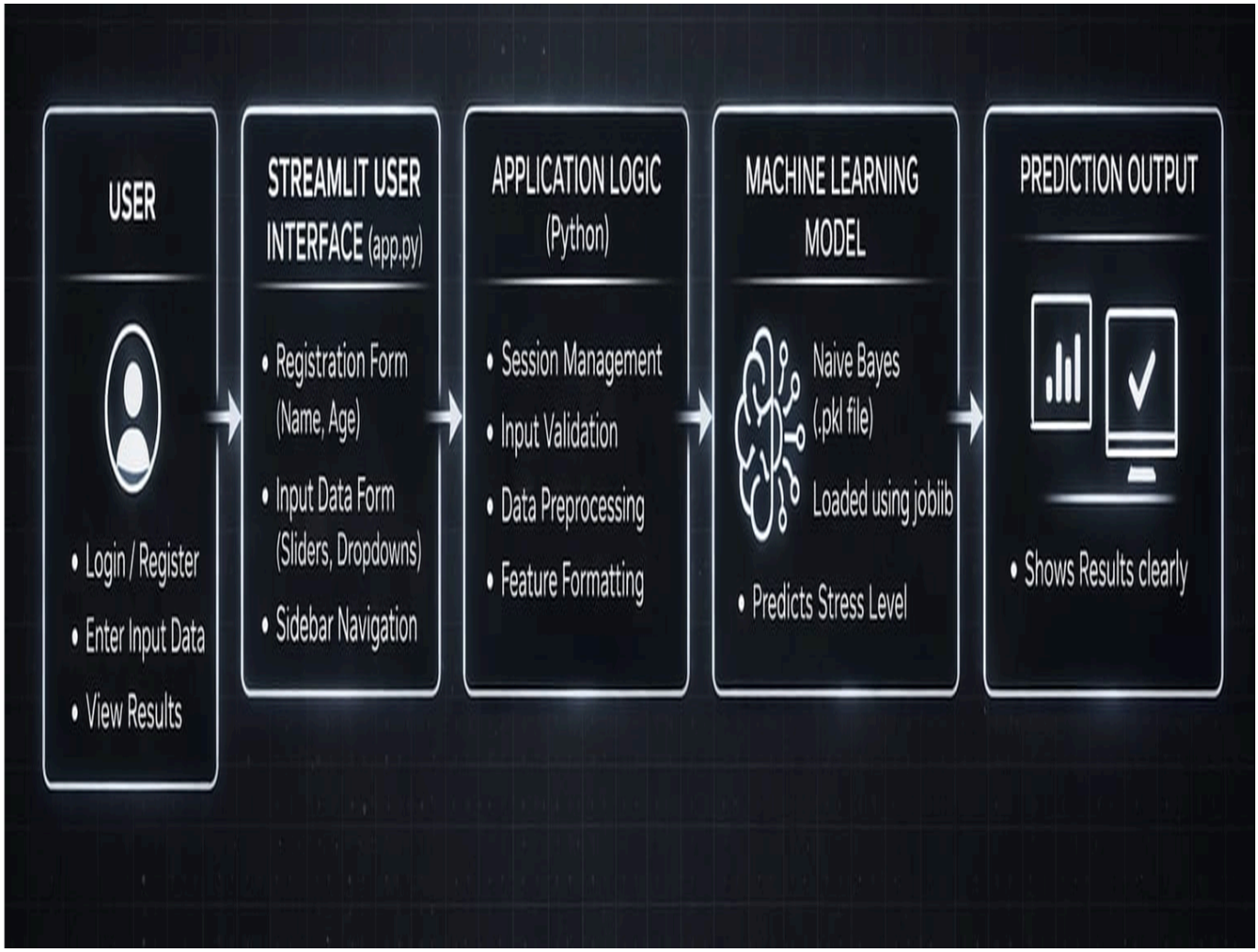


Fig 5.1: System architecture

5.2 UML DIAGRAMS

What is UML?

The Unified Modelling Language (UML) is a standard language for specifying, visualizing, constructing, and documenting the artifacts of software systems, as well as for business modelling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects. Using the UML helps project teams communicate, explore potential designs, and validate the architectural design of the software.

As the strategic value of software increases for many companies, the industry looks for techniques to automate the production of software and to improve quality and reduce cost and time-to-market. These techniques include component technology, visual programming, patterns and frameworks. Businesses also seek techniques to manage the complexity of systems as they increase in scope and scale. In particular, they recognize the need to solve recurring architectural problems, such as physical distribution, concurrency, replication, security, load balancing and fault tolerance. Additionally, the development for the World Wide Web, while making some things simpler, has exacerbated these architectural problems. The Unified Modelling Language (UML) was designed to respond to these needs. Simply, systems design refers to the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements, which can be done easily through UML diagrams.

Contents of UML:

In general, a UML diagram consists of the following features:

- ***Entities***

  These may be classes, objects, users, or systems behaviors.Relationship Lines that model the relationships between entities in the system.

- ***Generalization***

A solid line with an arrow that points to a higher abstraction of the present item.

- ***Association***

A solid line that represents that one entity uses another entity as part of its behaviour.

- ***Dependency***

A dotted line with an arrowhead that shows one entity depends on the behaviour of another entity.

Advantages:

- Provides standard for software development.
- Development time is reduced.
- The past faced issues by the developers are no longer exists.
- Has large visual elements to construct and easy to follow.
- To establish an explicit coupling between concepts and executable code.
- To creating a modeling language usable by both humans and machines UML defines several models for representing systems.

In this project six basic UML diagrams have been explained:

1. Class Diagram
2. Use Case Diagram
3. Sequence Diagram
4. Activity Diagram
5. Component Diagram
6. Deployment Diagram

5.2.1 Class Diagram

UML class diagrams are used to represent the static structure of a system by showing the relationships between different classes. These diagrams illustrate how the system is organized by identifying its main components, including classes, their attributes, methods, and interactions. Class diagrams are widely used in object-oriented

design as they provide a clear blueprint for system implementation and help in understanding the overall architecture.

In the **Mind Scope** system, the class diagram represents the core components involved in the mental health prediction process. It helps visualize how user input is processed, how predictions are generated, and how results and recommendations are provided.

The system consists of five main classes:

- **User Class:**
  Represents the user interacting with the system. It includes attributes such as username and password, along with methods like login() and logout() for authentication.
- **InputData Class:**
  Stores the input parameters provided by the user, such as anxiety level, depression level, sleep quality, study load, and social factors. This class acts as the data source for the prediction model.
- **PredictionModel Class:**
  Represents the machine learning component. It contains the Naive Bayes model and includes methods such as load_model() and predict(data) to process input data and generate predictions.
- **Result Class:**
  Stores the prediction output, including stress_level and confidence_score. It represents the final outcome of the system.
- **Recommendation Class:**
  Generates personalized suggestions based on the predicted stress level using the method generateSuggestions(stressLevel).

The relationships between these classes define the workflow of the system. The User provides input data to the InputData class, which is then passed to the PredictionModel

for analysis. The model generates a prediction, which is stored in the Result class. Based on this result, the Recommendation class is triggered to provide suitable suggestions.

Class diagrams support better system design by clearly defining responsibilities of each component and ensuring modularity. They also help in identifying potential design issues early and improve communication among developers and stakeholders.

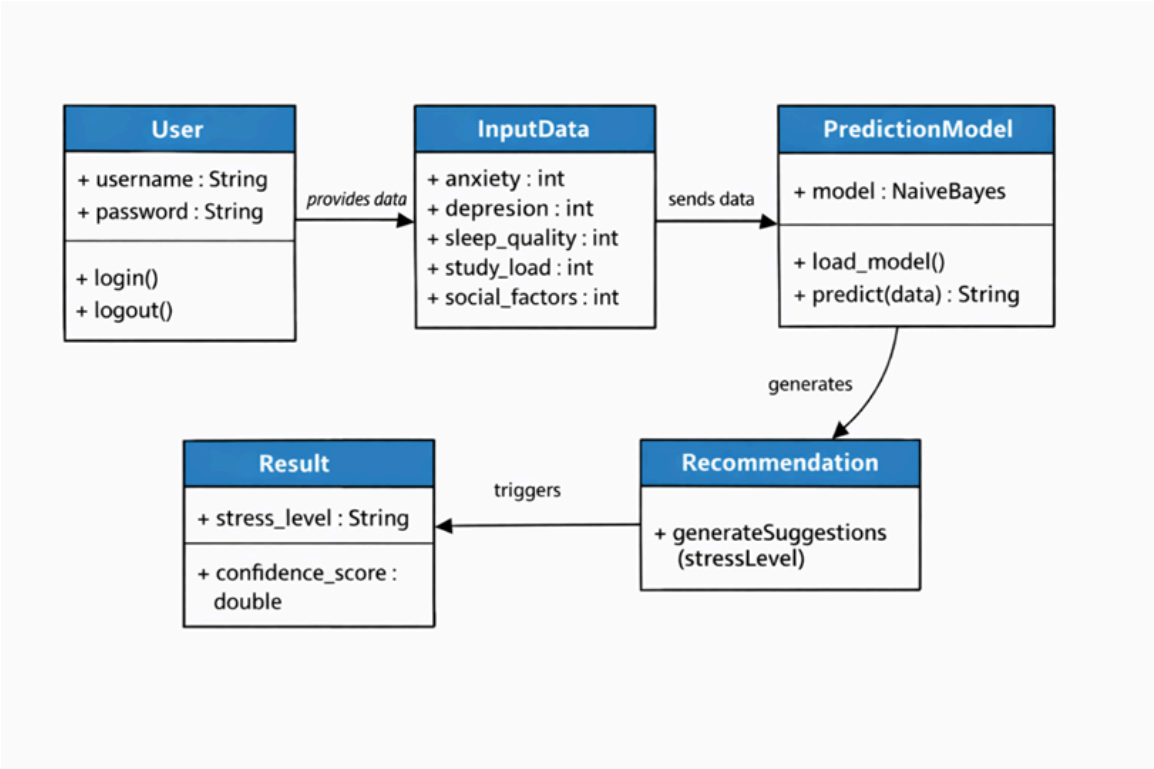


Fig 5.2.1: Class diagram

5.2.2 Use Case Diagram

A use case diagram is a behavioral UML diagram that represents the interactions between users (actors) and the system. It provides a high-level view of the functionalities offered by the system and helps in understanding how different users interact with various features.

In the Mind Scope system, the use case diagram illustrates the interaction between two main actors: User and Admin, and the functionalities provided by the system.

Actors in the System

- User:
  The primary actor who interacts with the system to analyze mental health.
- Admin:
  The secondary actor responsible for monitoring and analyzing system data.

Use Cases

The system provides the following functionalities:

- Login:
  Allows the user to access the system.
- Enter Input Data:
  Users provide mental health-related inputs such as anxiety, sleep quality, and lifestyle factors.
- Predict Stress Level:
  The system processes input data and predicts the stress level.
- View Results:
  Displays the predicted stress level to the user.
- View Recommendations:
  Provides suggestions based on the predicted stress level.
- View Analytics:
  Allows users or admin to analyze overall data trends.
- View Dataset Insights (Admin):
  Admin can access detailed dataset insights and analytics.

Relationships in the Diagram

- Association:

  The User is directly associated with all primary functionalities such as login, input, prediction, and viewing results.
- Include Relationship:

  The *View Analytics* use case includes *View Dataset Insights*, meaning analytics depends on dataset insights.
- Admin Interaction:

  Admin is connected to dataset insights for monitoring and analysis purposes.

Purpose of the Use Case Diagram

The use case diagram helps in:

- Understanding system functionalities at a high level
- Identifying interactions between users and the system
- Defining system requirements clearly
- Improving communication among developers and stakeholders

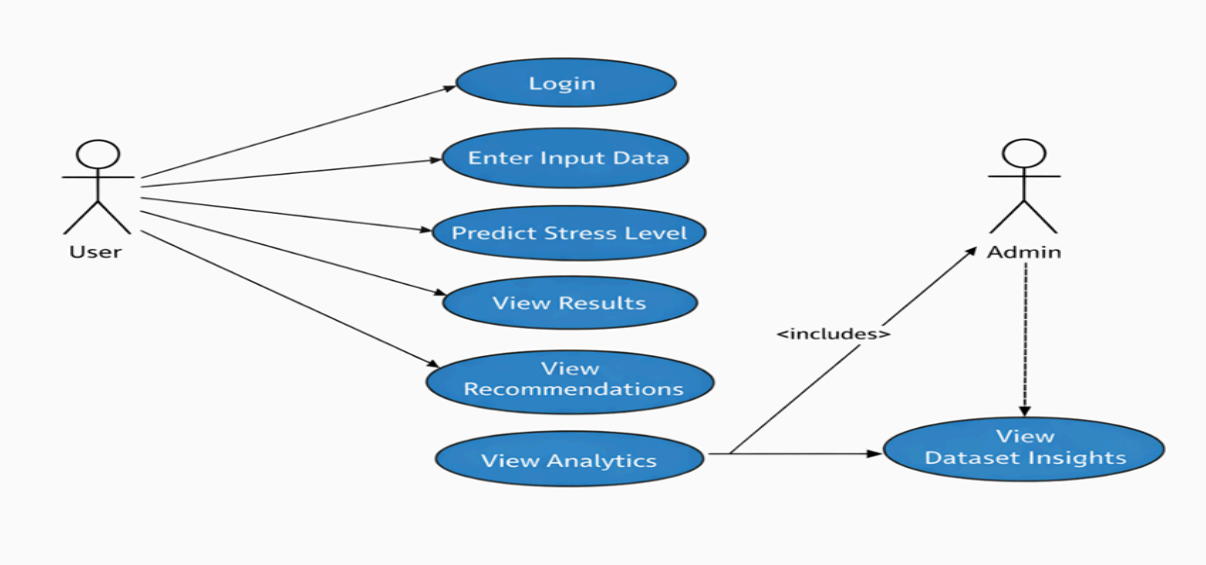


Fig 5.2.2: Use Case diagram

5.2.3 Sequence Diagram

A sequence diagram is a type of UML interaction diagram that represents how different components of a system interact with each other in a specific order over time. It helps in understanding the dynamic behavior of the system by showing how messages are passed between different modules to complete a process.

Components Involved

The main components in the system include User, InputData, PredictionModel, Result, and Recommendation. Each component plays a specific role in processing user data and generating predictions.

Working of the System

The process begins when the user logs into the system. After successful login, the user enters mental health-related data such as anxiety level, depression level, sleep quality, and study load. This data is collected and managed by the InputData module, which converts it into a structured format suitable for processing.

The processed data is then sent to the PredictionModel, where the Naive Bayes algorithm is applied to predict the stress level. The model analyzes the input data and generates the prediction result.

The generated result is returned and stored in the Result module, which contains the predicted stress level and confidence score. This result is then sent to the user interface for display.

At the same time, the Recommendation module is triggered based on the predicted stress level. It generates personalized suggestions to help improve the user's mental well-being.

Key Elements

The sequence diagram consists of lifelines representing system components and messages representing communication between them. The diagram clearly shows the order of execution, from user input to prediction and recommendation generation.

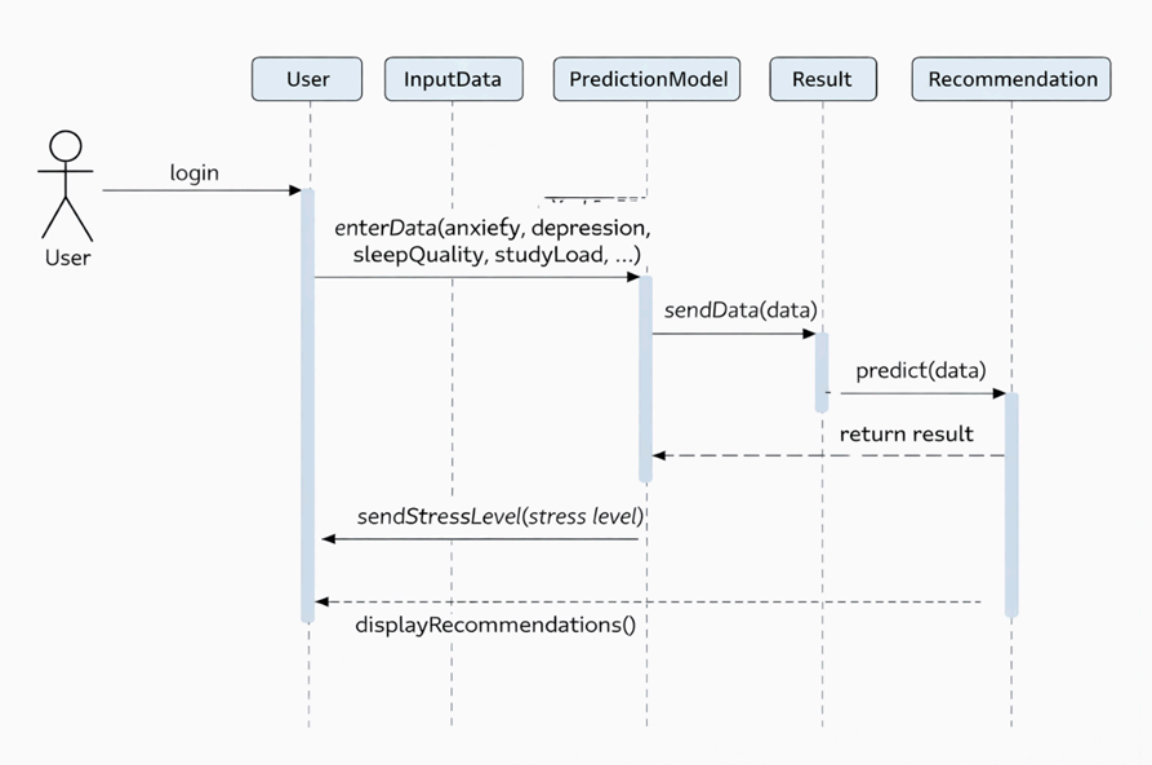


Fig 5.2.3: Sequence diagram

5.2.4 Activity Diagram

An activity diagram is a UML behavioral diagram used to represent the flow of activities within a system. It describes how different operations are carried out in a sequential manner, including conditions, decision points, and transitions between

activities. It helps in understanding the overall workflow of the system in a clear and simple way.

Overview of the System Flow

In the Mind Scope system, the activity diagram represents the complete flow of operations from user interaction to prediction and recommendation. It shows how the system processes user input and generates results step by step.

Working of the Activity Diagram

The process begins with the **Start** state, which indicates the initiation of the system. The user first performs the **Login** operation to access the system. After successful login, the user proceeds to **Enter Input Data**, where mental health-related parameters such as anxiety, sleep quality, and lifestyle factors are provided.

Once the data is entered, the system moves to the **Process Data** stage, where preprocessing and formatting of input data take place. The processed data is then used in the **Predict Stress Level** activity, where the Naive Bayes model analyzes the input and generates a prediction.

After prediction, a decision point occurs where the system determines whether the stress level is low or high. Based on this condition, the system proceeds to **Show Result**, displaying the predicted stress level to the user.

Following this, the system moves to the **Recommendations** stage, where personalized suggestions are generated to help improve mental well-being. Finally, the process reaches the **End** state, indicating completion of the workflow.

Key Elements in the Diagram

The activity diagram includes an initial state (start), multiple activity states such as login, input, processing, and prediction, a decision node for stress level classification, and a

final state (end). Arrows represent the flow of control between different activities, ensuring a clear understanding of the system's execution.

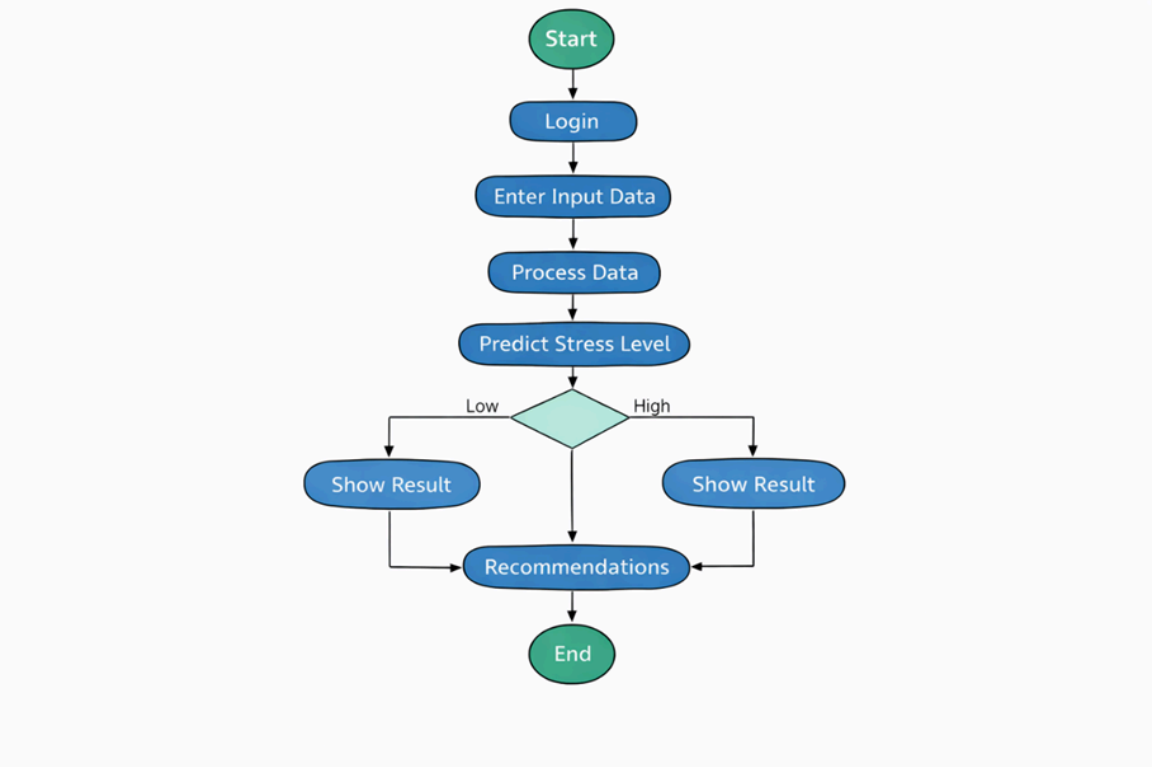


Fig 5.2.4: Activity diagram

5.2.5 Component Diagram

A component diagram is a UML structural diagram that represents the organization and interaction of different components within a system. It provides a high-level view of how various modules are connected and how data flows between them. This diagram helps in understanding the overall system architecture and ensures proper integration between components.

Overview of the System Components

In the Mind Scope system, the component diagram represents the major functional modules involved in mental health prediction. The system is divided into multiple components such as User Interface, Data Processing, Prediction Model, Analysis Module, and Recommendation Module. Each component performs a specific function and interacts with others to complete the workflow.

Working of the Component Diagram

The process begins with the User Interface component, which allows users to log in, enter input data, and view results. The user inputs mental health-related parameters such as anxiety, sleep quality, and lifestyle factors through this interface.

The input data is then passed to the Data Processing component, where it is prepared for prediction. This module handles data input, dataset handling, and receives processed stress levels from the prediction model.

The processed data is forwarded to the Prediction Model component, where the Naive Bayes algorithm is applied to predict the stress level. The model analyzes the dataset and sends the prediction results back to the system.

The predicted stress results are then sent to the Recommendation Module, which generates appropriate suggestions based on the stress level to help improve the user's mental well-being.

At the same time, the system includes an Analysis Module, which is responsible for providing insights and analytics based on the dataset. This module helps in understanding patterns and trends in mental health data.

Interaction Between Components

The components are connected through data flow relationships. The User Interface interacts with the Data Processing module by sending user input. The Data Processing module communicates with the Prediction Model for stress prediction. The Prediction Model sends results to both the Recommendation Module and Data Processing module. The Analysis Module interacts with the dataset to provide analytical insights.

Importance of the Component Diagram

The component diagram helps in visualizing the modular structure of the system and ensures clear separation of responsibilities among different components. It improves system design, scalability, and maintainability by clearly defining how each module interacts within the system.

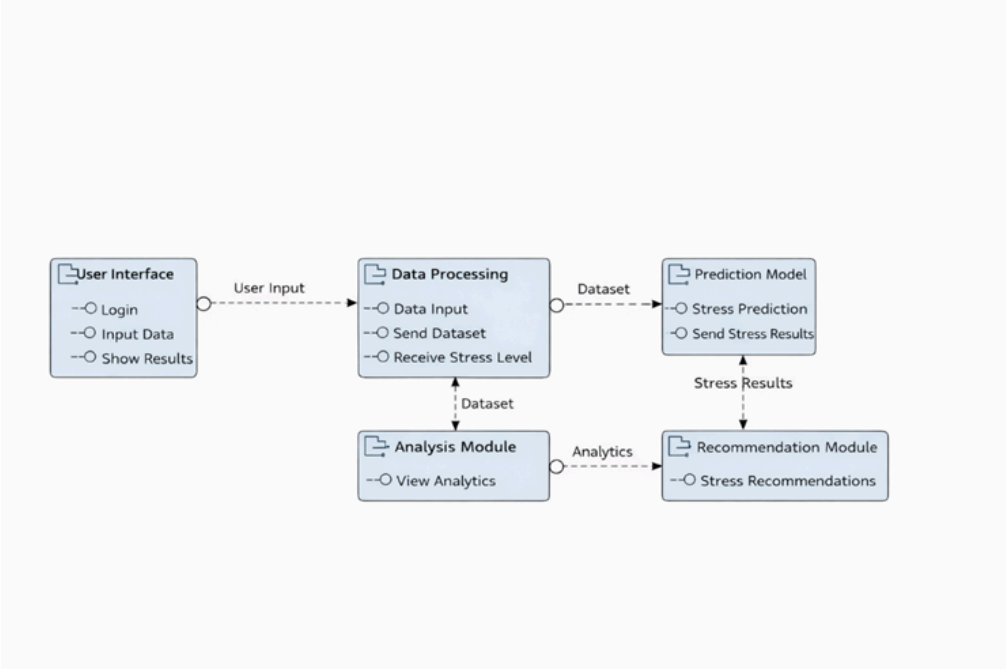


Fig 5.2.5: Component diagram

5.2.6 Deployment Diagram

A deployment diagram is a UML structural diagram that represents the physical deployment of software components on hardware systems. It shows how different parts of the application are distributed across devices such as servers, user systems, and platforms. This diagram helps in understanding the runtime environment and how the system is executed in real-world scenarios.

Overview of Deployment

In the Mind Scope system, the deployment diagram illustrates how the machine learning model and application are deployed and accessed by users through a web-based interface. The system is designed to run efficiently on local machines or cloud environments using a lightweight architecture.

System Deployment Architecture

The system consists of a **machine learning model**, which is trained using the Naive Bayes algorithm and stored as a serialized file. This model is integrated into a Python-based application developed using Streamlit.

The application can be deployed on a **local system or cloud server**, which acts as the main processing unit. The server handles user requests, processes input data, and interacts with the machine learning model to generate predictions.

Users access the system through a **web application interface**, where they can log in, enter input data, and view results. The interaction happens in real-time, and the system processes the data instantly without requiring complex infrastructure.

Working of the Deployment

When a user accesses the web application, the request is sent to the application server. The server processes the input using the integrated machine learning model and generates

the stress prediction. The result is then sent back to the user interface and displayed clearly along with recommendations.

The system does not require heavy cloud infrastructure or distributed systems, making it lightweight and easy to deploy. However, it can be extended to cloud platforms for scalability and wider accessibility.

Importance of the Deployment Diagram

The deployment diagram helps in visualizing how the system is physically implemented and accessed. It ensures proper understanding of system execution, improves deployment planning, and helps identify potential performance or scalability issues.

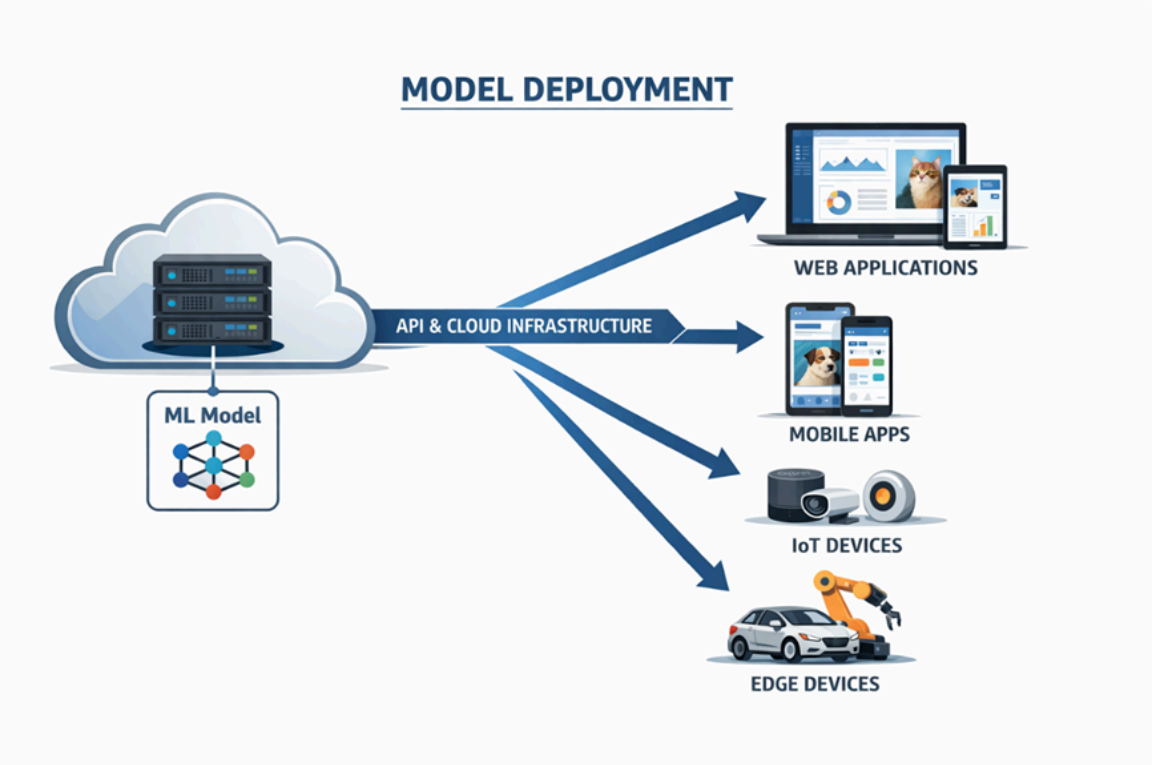


Fig 5.2.6: Deployment diagram

Chapter 6

SYSTEM IMPLEMENTATION

6.1 IMPLEMENTATION PROCESS

The implementation of the **Mind Scope: Mental Health Prediction System** is carried out through a structured sequence of steps, including data collection, preprocessing, model development, integration, and deployment. The system is designed to provide real-time prediction of mental health conditions using a machine learning approach.

Data Collection

The system uses a structured dataset containing mental health-related attributes such as anxiety levels, depression indicators, sleep quality, academic pressure, and social factors. The dataset is collected from publicly available sources and prepared for training the machine learning model.

Data Preprocessing

The collected data undergoes preprocessing to ensure quality and consistency. This includes handling missing values, encoding categorical variables, and converting data into a numerical format suitable for machine learning. Data normalization and cleaning are performed to improve model performance.

Model Selection and Training

The Naive Bayes algorithm is selected for this project due to its simplicity and efficiency in classification tasks. The model is trained using the prepared dataset, where it learns the relationship between input features and stress levels. The dataset is divided into training and testing sets to evaluate model performance.

Model Evaluation

After training, the model is evaluated using performance metrics such as accuracy, precision, recall, and F1-score. These metrics help in determining the reliability and effectiveness of the model in predicting mental health conditions.

System Development

The trained model is integrated into a web-based application developed using Streamlit. The application provides an interactive interface for users to input their data and receive predictions in real time.

Deployment

The system is deployed as a lightweight web application that can run on a local machine or a cloud platform. The deployment ensures that users can access the system easily and obtain instant results without requiring complex infrastructure.

6.2 MODULE IMPLEMENTATION

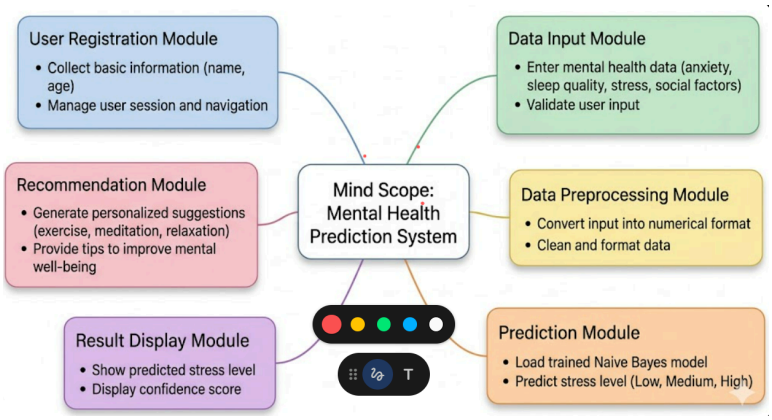


Fig 6.2.1 List of Modules

The Mind Scope system is developed using a modular architecture, where each module performs a specific function in the mental health prediction pipeline. This modular approach improves system organization, scalability, maintainability, and ease of debugging. Each module is designed to handle a distinct task, ensuring efficient data flow from user input to prediction and recommendation.

User Registration Module

The User Registration module is responsible for collecting and managing basic user information such as name and age. It enables personalized interaction with the system and helps maintain session continuity during usage. This module uses Streamlit's session state to temporarily store user data, ensuring that the user's details persist throughout the session without requiring a database.

The module also manages navigation within the application, allowing users to move between different sections such as input, prediction, and results. It ensures a smooth and

seamless user experience by maintaining the application state and preventing data loss during interaction.

Data Input Module

The Data Input module allows users to enter mental health-related information through an interactive and user-friendly interface. It collects key parameters such as anxiety level, depression level, sleep quality, study load, and social factors, which are essential for prediction.

The module utilizes Streamlit components such as sliders, dropdown menus, and input fields to make the data entry process intuitive and efficient. Input validation is also performed to ensure that the values entered fall within acceptable ranges, thereby improving data reliability and preventing incorrect predictions.

Data Preprocessing Module

The Data Preprocessing module is responsible for transforming raw user input into a structured format suitable for machine learning processing. It converts categorical inputs into numerical values, handles missing or inconsistent data, and ensures proper formatting of features.

This module also standardizes the input data based on the format used during model training. Feature alignment is performed to match the model's expected input structure. This step is crucial to ensure that the prediction model receives clean, consistent, and properly formatted data, which directly impacts prediction accuracy.

Prediction Module

The Prediction module is the core component of the system, where the actual machine learning computation takes place. This module loads the trained Naive Bayes model using Joblib and applies it to the processed input data.

The model calculates the probability of different stress levels based on input features and classifies the user's mental health condition into categories such as Low, Medium, or High. Due to the probabilistic nature of Naive Bayes, the module provides fast and efficient predictions, making it suitable for real-time applications.

The module ensures minimal latency and high responsiveness, allowing users to receive instant results upon submitting their data.

Result Display Module

The Result Display module is responsible for presenting the prediction output in a clear and user-friendly manner. It displays the predicted stress level along with additional indicators such as confidence score or probability percentage.

The module uses visual elements such as labels, indicators, and formatted text to enhance readability and user understanding. It ensures that the results are easily interpretable, even for non-technical users. This improves user engagement and helps users quickly understand their mental health status.

Recommendation Module

The Recommendation module generates personalized suggestions based on the predicted stress level. It provides actionable guidance such as engaging in physical activity, practicing meditation, improving sleep habits, and managing stress effectively.

The recommendations are designed to promote mental well-being and encourage users to take preventive or corrective measures. This module adds significant value to the system by not only identifying mental health conditions but also guiding users toward improvement.

Chapter 7

SYSTEM STUDY AND TESTING

7.1 FEASIBILITY STUDY

The feasibility study of the proposed **Mind Scope: Mental Health Prediction System** is conducted to evaluate whether the system is practical, efficient, and suitable for implementation. This analysis ensures that the system can be developed and deployed without placing excessive burden on resources.

Technical Feasibility

The technical feasibility evaluates whether the required technology and resources are available to develop the system. The Mind Scope system is implemented using Python, Streamlit, and machine learning libraries such as Scikit-learn, Pandas, and NumPy. These technologies are lightweight, widely available, and do not require high-end hardware. The system runs efficiently on standard computers, making it technically feasible and easy to implement.

Social Feasibility

Social feasibility determines the acceptance of the system by users. The proposed system is designed with a simple and user-friendly interface, allowing students and individuals to easily interact with it. Since the system helps in early detection of mental health conditions, it is expected to be well accepted by users. It promotes awareness and encourages individuals to monitor their mental well-being without hesitation.

Economic Feasibility

Economic feasibility evaluates the cost required to develop and maintain the system. The Mind Scope system uses open-source technologies such as Python and Streamlit, which

reduces development costs significantly. No expensive hardware or licensed software is required. Therefore, the system is cost-effective and can be developed within a limited budget.

7.2 SOFTWARE TESTING

Testing is an essential phase in the development of the MindScope Mental Health Prediction System. It ensures that the system functions correctly, meets user requirements, and produces accurate prediction results. The testing process involves validating each module individually as well as verifying the complete system functionality.

The system was tested using different types of testing techniques, including functional testing, unit testing, integration testing, and user acceptance testing. In functional testing, each feature such as login, data input, prediction, and recommendation was tested to ensure it performs the intended operation. The login module was tested with valid and invalid credentials to verify authentication accuracy. The data input module was tested by providing various combinations of inputs to ensure correct data collection and storage.

Unit testing was performed on individual modules such as the prediction module and data processing module. The machine learning model was tested using sample inputs from the dataset to verify that it produces expected outputs. Special attention was given to ensure that the model correctly classifies stress levels into Low, Medium, and High categories.

Integration testing was carried out to ensure smooth interaction between different modules. For example, data entered in the input module was successfully passed to the prediction module, and the results were correctly displayed in the result module. The recommendation module was also tested

to confirm that appropriate suggestions are generated based on prediction results.

The system was also tested for edge cases, such as minimum and maximum input values, to ensure stability. It was observed that the system handled these cases without errors. Performance testing was conducted to ensure that predictions are generated quickly without delays.

User acceptance testing was performed to evaluate the usability and user experience of the system. Users found the interface intuitive and easy to navigate. The visual representation of results, such as stress meters and color indicators, improved understanding.

Overall, the testing process confirmed that the system is reliable, efficient, and capable of providing accurate mental health predictions. The system meets all functional requirements and performs effectively under different conditions, ensuring readiness for real-world usage.

Test Case	Input	Expected Output	Result
Registration	Name, age data	Home page	Passed
Low Stress	Low values	Low Stress	Passed
High Stress	High values	High Stress	Passed

Table 7.2.1 System Validation And Testing Results

TEST CASE DESIGN

Test case design is an important part of the testing process, where different scenarios are created to validate the functionality and performance of the system.

In the MindScope Mental Health Prediction System, test cases are designed to ensure that each module behaves as expected under various conditions.

Each test case includes details such as test case ID, input conditions, expected output, and actual result. These test cases help in identifying errors and verifying system correctness. The system was tested with multiple input combinations, including normal values, boundary values, and extreme conditions.

For the login module, test cases were created to check valid and invalid user credentials. The system correctly allowed access for valid users and denied access for invalid credentials. For the input module, test cases ensured that all input fields accept valid ranges and reject incorrect values.

In the prediction module, test cases were designed using different combinations of input features such as low anxiety, high sleep quality, and low study load to verify low stress predictions. Similarly, high values of anxiety, depression, and peer pressure were tested to confirm high stress predictions. The system successfully classified stress levels in most cases.

Boundary testing was performed by providing minimum and maximum values for input features. The system handled these cases without crashing or producing incorrect outputs. This confirms system robustness and reliability.

Integration test cases verified that data flows correctly from input to prediction and then to the recommendation module. The system consistently displayed accurate results and relevant recommendations.

The following table shows some sample test cases used during testing:

Test Case ID	Module	Input Condition	Expected Output	Actual Output	Status
TC01	Registration	Enter name, age	Takes to home page	Takes to home page	Pass

TC03	Input	All fields filled	Data accepted	Data accepted	Pass
TC04	Prediction	Low values	Low stress	Low stress	Pass
TC05	Prediction	Medium values	Medium stress	Medium stress	Pass
TC06	Prediction	High values	High stress	High stress	Pass
TC07	Boundary	Min values (0)	Low stress	Low stress	Pass
TC08	Boundary	Max values (10/30)	High stress	High stress	Pass
TC09	Integration	Full flow	Correct output	Correct output	Pass
TC10	Recommendation	High stress	Suggestions shown	Suggestions shown	Pass

Table 7.2.2 Detailed Test Case Execution Log

UNIT TESTING

Unit testing is performed to verify the functionality of individual modules in isolation. In the MindScope system, each module such as login, data input, data processing, prediction, and recommendation was tested separately.

The login module was tested with different combinations of usernames and passwords to ensure correct authentication. The data input module was tested to verify that all input fields accept valid values within the specified range. The data processing module was tested to ensure that input values are properly scaled and transformed before being passed to the model.

The prediction module was tested using sample inputs to verify that it correctly classifies stress levels. The recommendation module was tested to ensure that appropriate suggestions are generated based on predicted stress levels.

Unit testing helped in identifying errors at an early stage and ensured that each module performs its intended function independently.

INTEGRATION TESTING

Integration testing is performed to verify the interaction between different modules of the system. In the MindScope system, integration testing ensured that data flows correctly from one module to another.

The data entered in the input module was successfully passed to the data processing module and then to the prediction module. The prediction results were correctly displayed in the result module, and appropriate recommendations were generated.

This testing confirmed that all modules are properly connected and work together as a complete system. It ensured that there are no data loss or communication errors between modules.

FUNCTIONAL TESTING

Functional testing is carried out to ensure that all features of the system work according to the specified requirements. Each functionality such as login, input data entry, prediction, result display, and recommendation was tested.

The login functionality was tested for both valid and invalid credentials. The prediction functionality was tested with various input combinations to ensure accurate classification of stress levels. The recommendation functionality was tested to verify that appropriate suggestions are displayed.

Functional testing ensured that all system features operate correctly and meet user requirements.

USER ACCEPTANCE TESTING

User Acceptance Testing is conducted to evaluate the system from the end-user perspective. In this phase, users interacted with the system to test its usability and effectiveness.

Users found the interface simple and easy to use. The prediction results were clear and understandable, and the recommendations were helpful. Feedback from users indicated that the system is useful for identifying mental health risks.

UAT ensured that the system satisfies user expectations and is ready for real-world deployment.

PERFORMANCE TESTING

Performance testing is conducted to evaluate the speed and efficiency of the system. The MindScope system was tested to ensure that predictions are generated quickly without delay.

The system showed fast response time even when handling multiple inputs. The model prediction process was efficient and required minimal computation time.

This testing confirmed that the system performs efficiently and provides quick results to users.

BOUNDARY VALUE TESTING

Boundary value testing is performed by providing minimum and maximum input values to check system behavior. Inputs such as anxiety level, sleep quality, and study load were tested with values like 0 and maximum limits.

The system handled these boundary conditions correctly and produced valid predictions without errors. This ensures that the system is robust and stable.

Boundary testing helps in identifying edge-case issues and improves system reliability.

VALIDATION TESTING

Boundary value testing is performed by providing minimum and maximum input values to check system behavior. Inputs such as anxiety level, sleep quality, and study load were tested with values like 0 and maximum limits.

The system handled these boundary conditions correctly and produced valid predictions without errors. This ensures that the system is robust and stable.

Boundary testing helps in identifying edge-case issues and improves system reliability.

Chapter 8

SOURCE CODE

```
import streamlit as st


st.set_page_config(page_title="MindScope Login", layout="centered")
```

```
# ---------- SESSION ----------
if "logged_in" not in st.session_state:
    st.session_state.logged_in = False

# ---------- IF ALREADY LOGGED IN ----------
if st.session_state.logged_in:
    st.success("Already logged in ✅")
    if st.button("Go to Home"):
        st.switch_page("pages/1_Home.py")
    st.stop()

# ---------- LOGIN UI ----------
st.title("🧠 MindScope")
st.subheader("Register")

name = st.text_input("Enter your name")
age = st.number_input("Enter your age", 1, 100)

if st.button("register"):
    if name:
        st.session_state.logged_in = True
        st.session_state.user_name = name

        st.success(f"Welcome {name} 👋")

        st.switch_page("pages/1_Home.py")
    else:
        st.warning("Please enter your name")
```

Input module

```
import streamlit as st

# ---------- LOGIN PROTECTION ----------
if "logged_in" not in st.session_state or not st.session_state.logged_in:
    st.warning("Please login first")
    st.switch_page("app.py")

st.set_page_config(layout="wide")

st.title("📊 Student Mental Health Input")

st.markdown("Provide the following information to analyze stress levels.")

st.image("images/student_input.png", use_container_width=True)

st.divider()

# ---------- HEALTH FACTORS ----------
st.subheader("🧠 Health Factors")

col1, col2 = st.columns(2)


with col1:
    anxiety = st.slider("Anxiety Level",0,10,5)
    depression = st.slider("Depression Level",0,10,5)
    headache = st.slider("Headache Frequency",0,10,5)
    sleep_quality = st.slider("Sleep Quality",0,10,5)

with col2:
    self_esteem = st.slider("Self Esteem",0,10,5)
```

```
    breathing_problem = st.slider("Breathing Problems",0,10,5)
    blood_pressure = st.slider("Blood Pressure Level",0,10,5)
    mental_health_history = st.selectbox("Mental Health History", ["No", "Yes"])

# Convert to model format
if mental_health_history == "Yes":
    mental_health_history = 1
else:
    mental_health_history = 0
st.divider()

# ---------- ENVIRONMENT FACTORS ----------
st.subheader("🏠 Environment Factors")

col1, col2 = st.columns(2)

with col1:
    noise_level = st.slider("Noise Level",0,10,5)
    living_conditions = st.slider("Living Conditions",0,10,5)
    safety = st.slider("Safety Level",0,10,5)

with col2:

   basic_needs = st.slider("Basic Needs Availability",0,10,5)

st.divider()

# ---------- ACADEMIC FACTORS ----------
st.subheader("📚 Academic Factors")

col1, col2 = st.columns(2)
```

```
with col1:
    academic_performance = st.slider("Academic Performance",0,10,5)
    study_load = st.slider("Study Load",0,10,5)

with col2:
    teacher_student_relationship = st.slider("Teacher Student Relationship",0,10,5)
    future_career_concerns = st.slider("Future Career Concerns",0,10,5)

st.divider()

# ---------- SOCIAL FACTORS ----------
st.subheader("👥 Social Factors")

col1, col2 = st.columns(2)

with col1:
    social_support = st.slider("Social Support",0,10,5)
    peer_pressure = st.slider("Peer Pressure",0,10,5)

with col2:
    extracurricular_activities = st.slider("Extracurricular Activities",0,10,5)

bullying = st.slider("Bullying Experience",0,10,5)

st.divider()

# ---------- PREDICT BUTTON ----------
if st.button("🚀 Predict Stress Level"):

    # Check if user changed any values
```

```
if (
    anxiety == 5 and depression == 5 and study_load == 5 and peer_pressure == 5
    and sleep_quality == 5 and social_support == 5
):
    st.warning("⚠ Please adjust the inputs before prediction.")
    st.stop()

st.session_state.input_data = [
    anxiety,
    self_esteem,
    mental_health_history,
    depression,
    headache,
    blood_pressure,
    sleep_quality,
    breathing_problem,
    noise_level,
    living_conditions,
    safety,
    basic_needs,
    academic_performance,
    study_load,

    teacher_student_relationship,
    future_career_concerns,
    social_support,
    peer_pressure,
    extracurricular_activities,
    bullying
]
```

```
    st.success("Student data saved successfully!")

    st.switch_page("pages/3_Prediction.py")

st.markdown("---")
st.caption("MindScope Mental Health Dashboard | Student Wellness System")

Prediction Module

import streamlit as st
import joblib
import numpy as np
import plotly.graph_objects as go
import plotly.express as px
import time

# ---------- LOGIN PROTECTION ----------
if "logged_in" not in st.session_state or not st.session_state.logged_in:
    st.warning("Please login first")
    st.switch_page("app.py")

st.set_page_config(layout="wide")

st.title("🤖 Stress Prediction Result")

# Fix numpy compatibility
np.float = float

# ---------- LOAD MODEL ----------
model = joblib.load("model/mental_health_model.pkl")
```

```
# ---------- CHECK INPUT ----------
if "input_data" not in st.session_state:

    st.warning("⚠ Please enter student data first in the Input Page.")

else:

    data = np.array([st.session_state.input_data])

    # ---------- LOADING SPINNER ----------
    with st.spinner("Analyzing student stress level..."):
        time.sleep(2)
        prediction = model.predict(data)[0]

    # ---------- PROBABILITY ----------
    probabilities = model.predict_proba(data)[0]

    low_prob = round(probabilities[0] * 100)
    medium_prob = round(probabilities[1] * 100)
    high_prob = round(probabilities[2] * 100)

    st.session_state.prediction = prediction

    # ---------- PROGRESS BAR ----------
    progress = st.progress(0)

    for i in range(100):
        time.sleep(0.01)
        progress.progress(i + 1)

    stress_labels = {
```

```
    0: "Low Stress",
    1: "Medium Stress",
    2: "High Stress"
}

st.divider()

# ---------- RESULT DISPLAY ----------
col1, col2 = st.columns([1,2])

with col1:

    st.subheader("Prediction Result")

    if prediction == 0:
        st.success("Low Stress Level Detected")

    elif prediction == 1:
        st.warning("Moderate Stress Level Detected")

    else:
        st.error("High Stress Level Detected")

    st.metric("Predicted Level", stress_labels[prediction])

# ---------- STRESS METER ----------
with col2:

    if prediction == 0:
        meter_value = 20
    elif prediction == 1:
```

```
        meter_value = 60
    else:
        meter_value = 90

    fig = go.Figure(go.Indicator(
        mode="gauge+number",
        value=meter_value,
        title={'text': "Stress Meter"},
        number={'suffix': "%"},
        gauge={
            'axis': {'range': [0,100]},
            'bar': {'color': "black"},
            'steps': [
                {'range': [0,33], 'color': "#2ecc71"},
                {'range': [33,66], 'color': "#f1c40f"},
                {'range': [66,100], 'color': "#e74c3c"}
            ],
            'threshold': {
                'line': {'color': "black", 'width': 4},
                'thickness': 0.75,
                'value': meter_value
            }

        }
    ))

    st.plotly_chart(fig, use_container_width=True)

st.divider()

# ---------- STRESS PROBABILITY ----------
```

```
st.subheader("Stress Prediction Probability")

col1, col2, col3 = st.columns(3)

with col1:
    st.metric("Low Stress", f"{low_prob}%")

with col2:
    st.metric("Medium Stress", f"{medium_prob}%")

with col3:
    st.metric("High Stress", f"{high_prob}%")

# ---------- PROBABILITY CHART ----------
prob_data = {
    "Stress Level": ["Low", "Medium", "High"],
    "Probability": [low_prob, medium_prob, high_prob]
}

fig2 = px.bar(
    prob_data,
    x="Stress Level",

     y="Probability",
    color="Stress Level",
    title="Stress Prediction Probability"
)

st.plotly_chart(fig2, use_container_width=True)

st.divider()
```

```
    # ---------- STRESS ANALYSIS ----------
    st.subheader("Stress Analysis")

    if prediction == 0:

        st.info("""
The student currently shows **low stress indicators**.

Mental health condition appears stable.
Continue maintaining a balanced lifestyle and healthy study habits.
""")

    elif prediction == 1:

        st.warning("""
The student shows **moderate stress indicators**.

Academic workload and lifestyle factors may be contributing.
Early stress management techniques are recommended.
""")


   else:

        st.error("""
The student shows **high stress indicators**.

Immediate attention is recommended including counseling
and lifestyle adjustments.
""")
```

```
    st.divider()

    # ---------- NAVIGATION ----------
    col1, col2 = st.columns(2)

    with col1:
        if st.button("🔄 Start New Prediction"):
            del st.session_state["input_data"]
            st.switch_page("pages/2_Input_Data.py")

    with col2:
        if st.button("💡 View Recommendations"):
            st.switch_page("pages/4_Recommendations.py")

st.markdown("---")
st.caption("MindScope Mental Health Dashboard | Student Wellness System")
```

Recommendation Module

```
import streamlit as st


# ---------- LOGIN PROTECTION ----------
if "logged_in" not in st.session_state or not st.session_state.logged_in:
    st.warning("Please login first")
    st.switch_page("app.py")

st.set_page_config(layout="wide")

st.title("💡 Mental Health Recommendations")
```

```
# ---------- STEP TRACKER ----------
st.progress(1.0)
st.markdown("### Step 3️⃣ : Mental Health Recommendations")

st.image("images/wellness_recommendation.jpg", use_container_width=True)

st.divider()

# ---------- CHECK PREDICTION ----------
if "prediction" not in st.session_state:

    st.warning("⚠️ Please complete the prediction step first.")

else:

    level = st.session_state.prediction

    col1, col2, col3 = st.columns(3)

    # ---------- LOW STRESS ----------
    if level == 0:

        st.success("Low Stress Level")

        with col1:
            st.info("🧘 Maintain Healthy Lifestyle")
            st.write("""
- Continue regular exercise
- Maintain a balanced diet
- Practice relaxation techniques
```

```
""")

        with col2:
            st.info("👥 Stay Socially Connected")
            st.write("""
- Spend time with friends and family
- Participate in social activities
- Maintain positive relationships
""")

        with col3:
            st.info("📚 Balanced Study Routine")
            st.write("""
- Maintain effective study habits
- Take regular study breaks
- Balance study and leisure time
""")

    # ---------- MEDIUM STRESS ----------
    elif level == 1:

        st.warning("Moderate Stress Level")

        with col1:
            st.info("😴 Improve Sleep Quality")
            st.write("""
- Aim for 7–8 hours of sleep
- Avoid late-night screen time
- Maintain a regular sleep schedule
""")
```

```
        with col2:
            st.info("🧠 Manage Stress")
            st.write("""
- Practice deep breathing
- Try meditation or mindfulness
- Plan daily activities effectively
""")

        with col3:
            st.info("📖 Reduce Academic Pressure")
            st.write("""
- Break large tasks into smaller goals
- Ask for help when needed
- Maintain healthy study habits
""")

    # ---------- HIGH STRESS ----------
    else:

        st.error("High Stress Level")

        with col1:

            st.info("👨‍⚕️ Seek Professional Support")
            st.write("""
- Speak with a counselor
- Consult mental health professionals
- Seek guidance from trusted mentors
""")

        with col2:
```

```
            st.info("🧘 Practice Relaxation")
            st.write("""
- Engage in yoga or meditation
- Spend time outdoors
- Practice calming breathing exercises
""")

        with col3:
            st.info("❤️ Emotional Support")
            st.write("""
- Talk openly with family or friends
- Build a strong support network
- Avoid isolation during stressful times
""")

    st.divider()

    # ---------- NAVIGATION ----------
    col1, col2 = st.columns(2)

    with col1:
        if st.button("🏠 Back to Home"):

            st.switch_page("pages/1_Home.py")

    with col2:
        if st.button("🔄 Start New Prediction"):

            if "input_data" in st.session_state:
                del st.session_state["input_data"]
```

```
        if "prediction" in st.session_state:
            del st.session_state["prediction"]

        st.switch_page("pages/2_Input_Data.py")

st.divider()

# ---------- FINAL MESSAGE ----------
st.markdown("""
### 🌿 Wellbeing Reminder

Maintaining mental wellbeing is important for overall health.
Developing positive habits, seeking support when needed, and maintaining balance in daily life can help reduce stress.
""")

st.caption("MindScope Mental Health Dashboard | Student Wellness System")
```

Model Training Code

```
import pandas as pd
from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score
import joblib

# Load dataset
data = pd.read_csv("StressLevelDataset.csv")

# Features and target
```

```
X = data.drop("stress_level", axis=1)   # change if column name is different
y = data["stress_level"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Naive Bayes model
model = GaussianNB()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Accuracy
acc = accuracy_score(y_test, y_pred)
print("Naive Bayes Accuracy:", acc)

# Save model
joblib.dump(model, "model/naive_bayes_model.pkl")
```

Chapter 9

SCREEN LAYOUTS

9.1 INTERFACE LAYOUT:

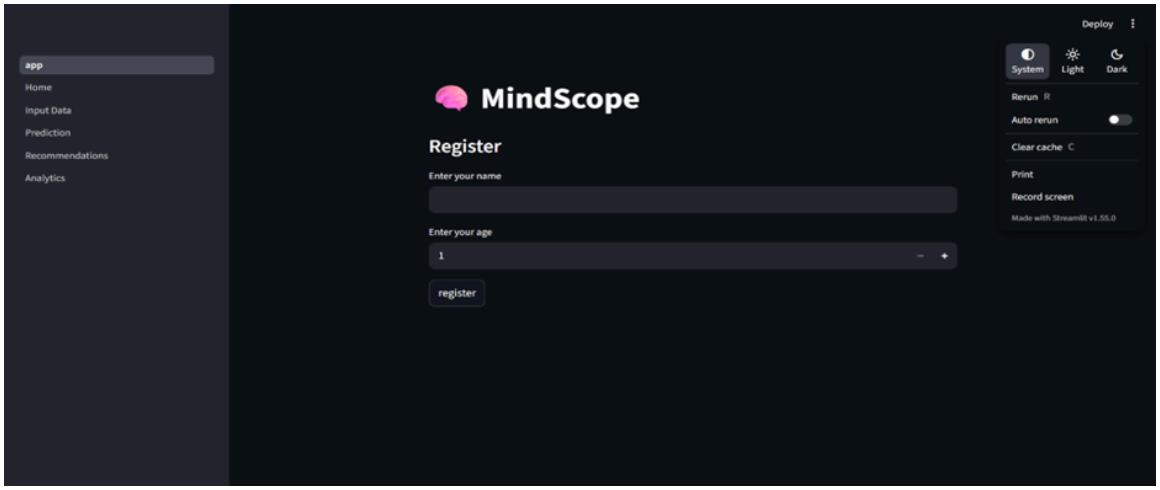


Fig 9.1 Interface layout

9.1.1 Home page:



Fig 9.1.1 home page

9.1.2 Input:



Fig 9.1.2 input Interface

9.1.3 input data

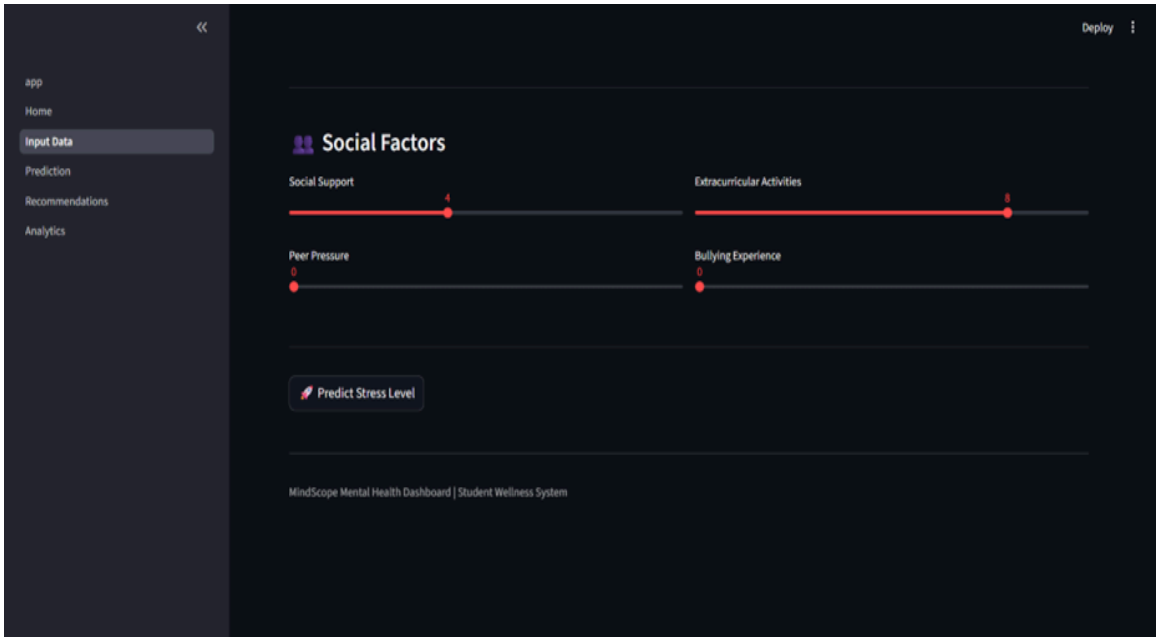


Fig 9.1.3 input data

9.2 RESULT:

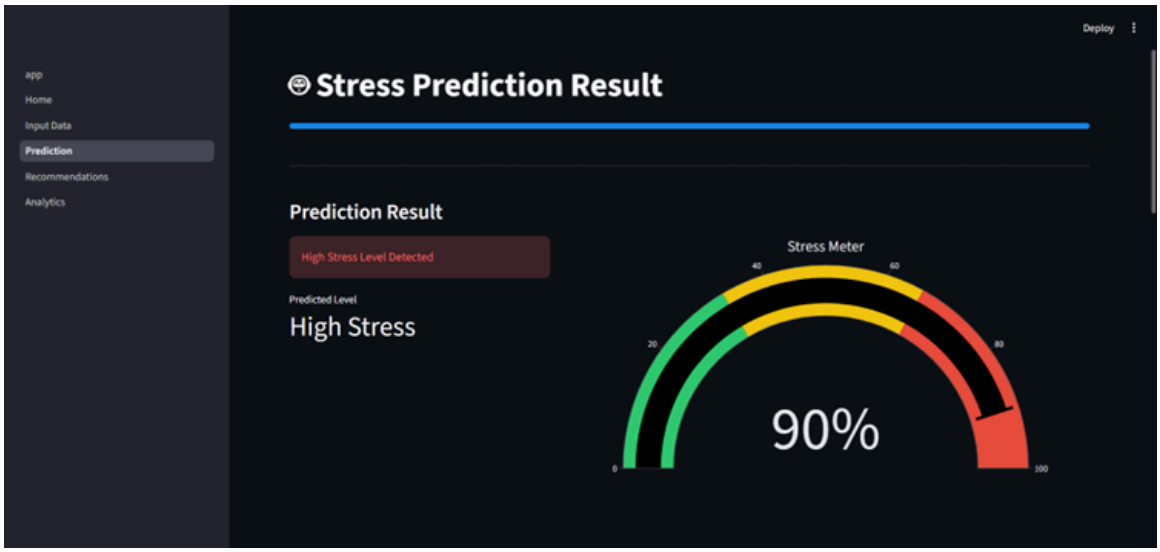


Fig 9.2.1 Output 1

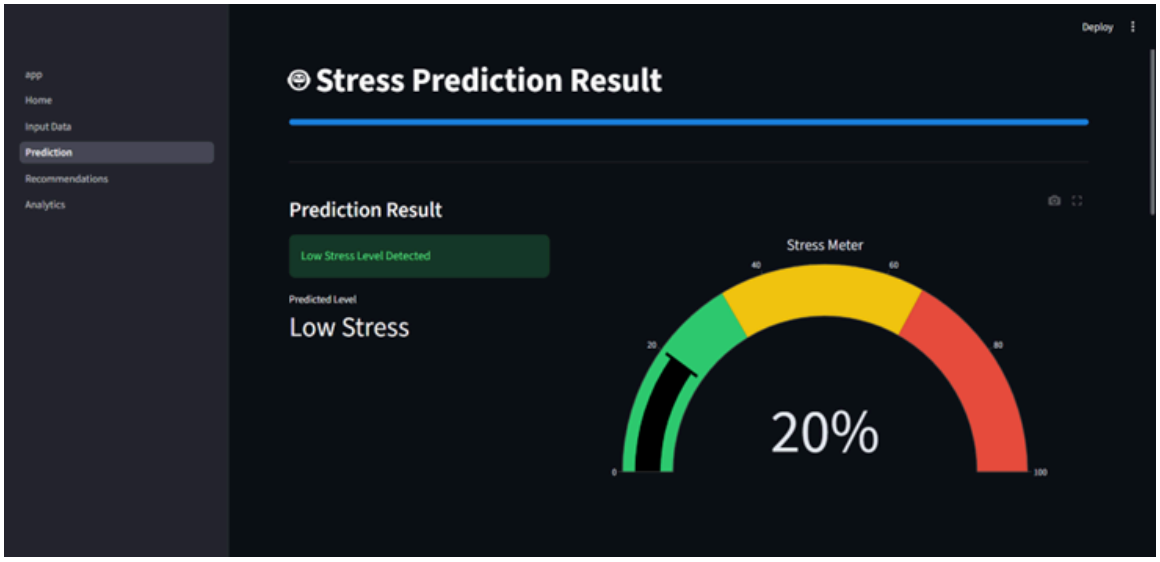


Fig 9.2.2 Output 2

Chapter 10

FUTURE ENHANCEMENTS

Future enhancements for the Mind Scope: Mental Health Prediction System aim to improve prediction accuracy, user experience, and real-world applicability. The current system can be extended by incorporating more advanced machine learning techniques such as ensemble models and deep learning algorithms to enhance prediction performance and handle more complex patterns in mental health data.

The system can be improved by integrating real-time data collection through wearable devices and mobile sensors to monitor parameters such as sleep patterns, physical activity, and heart rate.

Another enhancement includes expanding the dataset by incorporating data from diverse populations and real-world surveys, which will improve the generalization capability of the model.

The application can be extended into a mobile-based platform, allowing users to access mental health predictions anytime and anywhere. Integration with cloud services can improve scalability and enable large-scale usage.

The recommendation system can be further enhanced by incorporating AI-based personalized suggestions, including adaptive coping strategies, guided meditation, and mental health resources based on user behavior and history.

Future versions of the system can also include chatbot integration for interactive mental health support, enabling users to communicate and receive guidance in real time. Additionally, incorporating data visualization dashboards and analytics can help users and professionals better understand mental health trends.

Overall, these enhancements will make the system more intelligent, scalable, and effective in providing early detection and support for mental health conditions.

Chapter 11

CONCLUSION

This project presents an intelligent and user-friendly **Mental Health Prediction System**, titled *Mind Scope*, which utilizes machine learning techniques to predict stress levels among individuals. The system addresses the growing concern of mental health issues by providing an efficient and accessible solution for early detection and awareness.

The proposed system uses the **Naive Bayes algorithm** to analyze various factors such as anxiety levels, sleep quality, academic pressure, and social conditions. The model is trained on a structured dataset, and preprocessing techniques are applied to ensure data quality and improve prediction accuracy. The performance of the model is evaluated using standard metrics, demonstrating reliable and fast predictions suitable for real-time applications.

A web-based application is developed using **Streamlit**, providing an interactive interface where users can input their data and instantly receive predictions along with meaningful recommendations. The system is lightweight, easy to use, and does not require complex infrastructure, making it suitable for students and general users.

Overall, the Mind Scope system offers an effective solution for predicting mental health conditions and promoting awareness. It has the potential to assist individuals, educators, and healthcare professionals in identifying stress levels early and taking appropriate actions to improve mental well-being.

CHAPTER 12

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